

#HW 8

● Graded

12 Hours, 30 Minutes Late

Student

Daohn Kim

Total Points

86 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

6 / 10 pts

- 0 pts Correct

✓ - 4 pts $\Delta M = \Delta t (M_{in} - M_{out})$

- 2 pts the calculation was asked for 1 hour, not 30 mins.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $m_1 = m_2 + m_3$

- 3 pts $m_1 = 10.27 \text{ kg/s}$, $m_2 = m_3 = 5.13 \text{ kg/s}$

- 5 pts $1 \text{ bar} = 10^5 \text{ Pa}$; $m_1 = m_2 + m_3$ where, $m_2 = m_3$.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $Q = m [C_p \Delta T + (V_2^2 - V_1^2)/2]$

- 5 pts rate of heat transfer not calculated/ not included in scan

- 4 pts $v_1 = 6.25 \text{ m/s}$; $v_2 = 7.37 \text{ m/s}$; $Q = 10.56 \text{ kW}$

- 2 pts $v_2 = 7.37 \text{ m/s}$

Question 5

(no title)

10 / 15 pts

– 0 pts Correct

✓ – 5 pts $v_2 = \sim 149 \text{ ft/s}$; check the unit conversions carefully.

– 7 pts incomplete execution of solution

Question 6

(no title)

10 / 15 pts

– 0 pts Correct

– 3 pts 133.1 kW

✓ – 5 pts 133.1 kW; check the calculation clearly.

No questions assigned to the following page.

CHE 201 HW 8

Due: 11.59pm on 10/27/2025

1. (30 points)

Zybooks

Participation points

2. (10 points)

Problem 7.18.6

3. (15 points)

Problem 7.18.14

4. (15 points)

Problem 7.18.18

5. (15 points)

Problem 7.18.25

6. (15 points)

Problem 7.18.31

Question assigned to the following page: [2](#)

2.

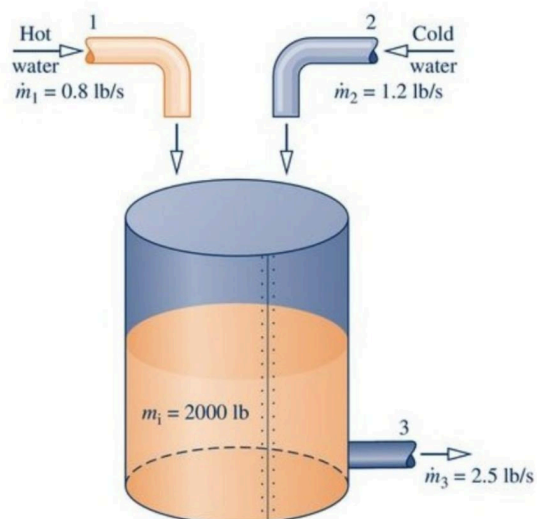


EXERCISE

7.18.6: (Problem 4.6 in the 9th edition).



- (a) The figure below shows a mixing tank initially containing 2000 lb of liquid water. The tank is fitted with two inlet pipes, one delivering hot water at a mass flow rate of 0.8 lb/s and the other delivering cold water at a mass flow rate of 1.2 lb/s. Water exits through a single exit pipe at a mass flow rate of 2.5 lb/s. Determine the amount of water, in lb, in the tank after one hour.



Feedback?

$$\dot{m}_i = \dot{m}_1 + \dot{m}_2 = 0.8 \text{ lb/s} + 1.2 \text{ lb/s} = 2.0 \text{ lb/s}$$

$$\dot{m}_e = \dot{m}_3 = 2.5 \text{ lb/s}$$

$$\Delta m = \Delta t (\dot{m}_e - \dot{m}_i) = 1 \text{ h} \cdot \frac{60 \text{ min}}{1 \text{ h}} \cdot \frac{60 \text{ s}}{1 \text{ min}} \cdot (2.5 \text{ lb/s} - 2.0 \text{ lb/s})$$

$$= 1800 \text{ lb}$$

$$m + \Delta m = 2000 \text{ lb} + 1800 \text{ lb} = \boxed{3800 \text{ lb}}$$

Question assigned to the following page: [3](#)

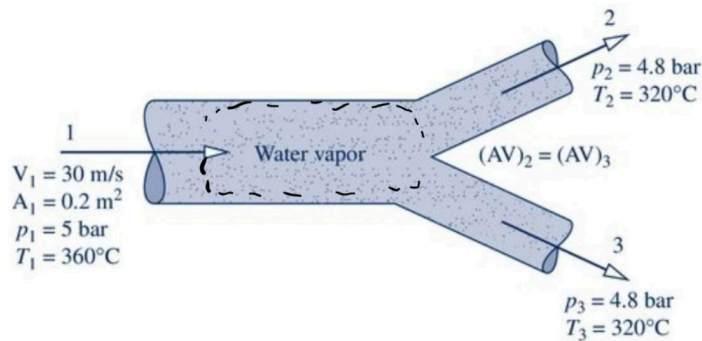
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EXERCISE

7.18.14: (Problem 4.14 in the 9th edition).



- (a) The figure below provides steady-state data for water vapor flowing through a piping configuration. At each exit, the volumetric flow rate, pressure, and temperature are equal. Determine the mass flow rate at the inlet and exits, each in kg/s.



→ steady state

Feedback?

M.B

$$\sum \dot{m}_i = \sum \dot{m}_e \quad (5.5)$$

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3$$

$$\frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} + \frac{A_3 V_3}{v_3}$$

$$v_1 \Rightarrow 5 \text{ bar } 360^\circ\text{C}, \quad 0.5796 \text{ m}^3/\text{kg}$$

$$v_2 \Rightarrow 4.8 \text{ bar } 320^\circ\text{C},$$

$$\text{at } 320^\circ\text{C}: \quad 3 \text{ bar} \quad 4.8 \text{ bar} \quad 5 \text{ bar}$$

$$v \quad 0.907$$

$$0.5416 \text{ m}^3/\text{kg}$$

$$\frac{0.5416 - 0.907}{5 - 3} = \frac{v_2 - 0.907}{4.8 - 3}$$

$$v_2 = 0.5781 \text{ m}^3/\text{kg}$$

$$v_3 = v_2 \quad (\text{in the same conditions})$$

$$\dot{m}_1 = \frac{A_1 V_1}{v_1} = \frac{0.2 \text{ m}^2 \cdot 30 \text{ m/s}}{0.5796 \text{ m}^3/\text{kg}} = \boxed{10.35 \text{ kg/s}}$$

$$\dot{m}_1 = 2 \cdot \left(\frac{A_2 V_2}{v_2} \right) \quad (\because A_2 V_2 = A_3 V_3, \quad v_2 = v_3)$$

$$= 2 \dot{m}_2 \quad \dot{m}_2 = \boxed{5.18 \text{ kg/s}}$$

$$\dot{m}_3 = \dot{m}_2 = \boxed{5.18 \text{ kg/s}}$$

Question assigned to the following page: [4](#)

3.



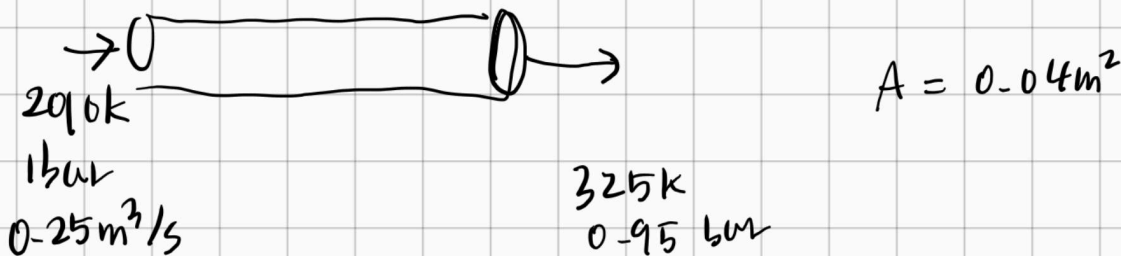
EXERCISE

7.18.18: (Problem 4.18 in the 9th edition).



- (a) Air enters a horizontal, constant-diameter heating duct operating at steady state at 290 K, 1 bar, with a volumetric flow rate of 0.25 m³/s, and exits at 325 K, 0.95 bar. The flow area is 0.04 m². Assuming the ideal gas model with $k = 1.4$ for the air, determine (a) the mass flow rate, in kg/s, (b) the velocity at the inlet and exit, each in m/s, and (c) the rate of heat transfer, in kW.

Feedback?



a) 1 bar, 290 K, $h = 290.16\text{ kJ/kg}$, $u = 206.91\text{ kJ/kg}$

$$h = u + p v \quad v = \frac{h - u}{p} = \frac{(290.16\text{ kJ/kg} - 206.91\text{ kJ/kg})}{(1\text{ bar})\left(\frac{10^2\text{ kJ}}{1\text{ bar}\cdot\text{m}^3}\right)} = 0.8325\text{ m}^3/\text{kg}$$

$$\dot{m}_i = \frac{0.25\text{ m}^3/\text{s}}{0.8325\text{ m}^3/\text{kg}} = 0.300\text{ kg/s}$$

at steady state, $\dot{m}_i = \dot{m}_e = 0.300\text{ kg/s}$

b) $\dot{V}_i = A_i V_i \quad V_i = \frac{\dot{V}_i}{A_i} = \frac{0.25\text{ m}^3/\text{s}}{0.04\text{ m}^2} = 6.25\text{ m/s}$

$$\dot{m}_e = \frac{A_e \bar{V}_e}{v_e}$$

325 K, $h = 325.31\text{ kJ/kg}$, $u = 232.02\text{ kJ/kg}$

$$v_e = \frac{h_e - u_e}{p_e} = \frac{(325.31\text{ kJ/kg} - 232.02\text{ kJ/kg})}{(0.95\text{ bar})\left(\frac{10^2\text{ kJ}}{1\text{ bar}\cdot\text{m}^3}\right)} = 0.982\text{ m}^3/\text{kg}$$

$$\bar{V}_e = \dot{m}_e \cdot \frac{v_e}{A_e} = (0.300\text{ kg/s}) \frac{(0.982\text{ m}^3/\text{kg})}{(0.04\text{ m}^2)} = 7.365\text{ m/s}$$

c) $\frac{E_{\text{in}}}{dt} = \dot{Q} - \dot{W}_{\text{cv}} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$

$$\dot{Q} = \dot{m}_i (h_e - h_i) = (0.300\text{ kg/s}) (325.31\text{ kJ/kg} - 290.16\text{ kJ/kg})$$

$$= 10.545 \text{ kJ/s}$$

Question assigned to the following page: [5](#)

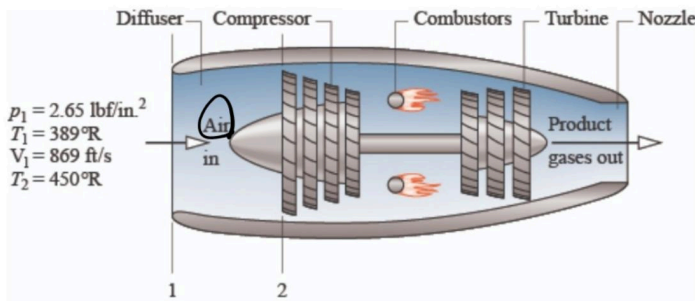
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EXERCISE

7.18.25: (Problem 4.25 in the 9th edition).



- (a) As shown in the figure below, air enters the diffuser of a jet engine operating at steady state at 2.65 lbf/in^2 , 389°R and a velocity of 869 ft/s , all data corresponding to high-altitude flight. The air flows adiabatically through the diffuser and achieves a temperature of 450°R at the diffuser exit. Using the ideal gas model for air, determine the velocity of the air at the diffuser exit, in ft/s .

 V_e

Engineering model

- diffuser
- steady state
- ideal gas
- adiabatic

$$M.B: \dot{m}_u = \dot{m}_e$$

$$E.B: \frac{E_{in}}{dt} = \dot{Q} - \dot{W}_{cv} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

$$\dot{m}_i \left(h_i + \frac{V_i^2}{2} \right) = \dot{m}_e \left(h_e + \frac{V_e^2}{2} \right)$$

 389°R

$$h_i = 389^\circ\text{R} \quad 90.75$$

 400°R

$$95.53 \text{ (Btu/lb)}$$

$$\frac{95.53 - 90.75}{400 - 389} = \frac{h_i - 90.75}{389 - 389}$$

$$h_i = 92.90 \text{ Btu/lb}$$

 450°R

$$h_e = 450^\circ\text{R} \quad 105.11$$

 460°R

$$109.90 \text{ (Btu/lb)}$$

$$\frac{109.90 - 105.11}{460 - 450} = \frac{h_e - 105.11}{450 - 450}$$

$$h_e = 107.51 \text{ Btu/lb}$$

from E.B

$$h_i + \frac{V_i^2}{2} = h_e + \frac{V_e^2}{2}$$

$$V_e^2 = 2(h_i - h_e) + V_i^2$$

$$= 2(92.90 - 107.51 \text{ Btu/lb}) + (869 \text{ ft/s})^2$$

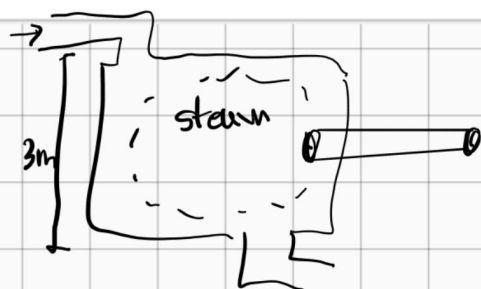
$$= 755131.78$$

$$V_e = \underline{868.98 \text{ ft/s}}$$

Question assigned to the following page: [6](#)

- (a) Steam enters a turbine operating at steady state with a mass flow of 10 kg/min, a specific enthalpy of 3100 kJ/kg, and a velocity of 30 m/s. At the exit, the specific enthalpy is 2300 kJ/kg and the velocity is 45 m/s. The elevation of the inlet is 3 m higher than at the exit. Heat transfer from the turbine to its surroundings occurs at a rate of 1.1 kJ per kg of steam flowing. Let $g = 9.81 \text{ m/s}^2$. Determine the power developed by the turbine, in kW.

Feedback?



model engineering

→ turbine

→ steady state

$$\dot{m}_i = \dot{m}_e = 10 \text{ kg/min} = 10 \text{ kg/min} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \frac{1}{6} \text{ kg/s}$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}_{cv} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

$$\dot{Q} = -11 \text{ kJ/kg}$$

$$z_i - z_e = 3 \text{ m}$$

$$0 = -11 \text{ kJ/kg} - \dot{W}_{cv} + \frac{1}{6} \text{ kg/s} (3100 \text{ kJ/kg} - 2300 \text{ kJ/kg}) + \frac{1}{6} \text{ kg/s} \left(\frac{(30 \text{ m/s})^2}{2} - \frac{(45 \text{ m/s})^2}{2} \right) + \frac{1}{6} (9.81 \text{ m/s}^2) (3 \text{ m})$$

$$\begin{aligned} \dot{W}_{cv} &= -11 \text{ kJ/kg} + \frac{1}{6} \text{ kg/s} (800 \text{ kJ/kg} - 562.5 \text{ kJ/kg} + 29.43 \text{ kJ/kg}) \\ &= 33.5 \text{ kJ/kg} \\ &= 33.5 \text{ kW} \end{aligned}$$

